The Very Standard Tricks in Algebra

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Contents

1 Inequalities

There are in general two types of inequality problems in the Olympiad. One is to find the optimum values (maximum and/or minimum) of an expression, usually a function in several variables (could be n variables in general) which are real numbers, or in most cases positive real numbers. In doing such problems, say, minimization problems, we need to find a lower bound for the expression and prove that the lower bound can be attained (e.g. when $x = y = 1$). It is a common mistake to provide a lower bound only, but the fact is that this lower bound cannot be attained and this is not the meaning of 'minimum'.

The other one is to prove an inequality LHS $\geqslant$ RHS where both sides are expressions in some variables. Here, to 'prove' means that we need to show that the inequality is true for any choice of the variables. Since there are always infinitely many possible choices of the variables, we cannot prove the inequality simply by substituting some values to the variables. Even if we are asked to prove an inequality only, it is better to write down the equality cases (the cases when LHS = RHS). Indeed, sometimes finding the equality cases might help us deduce a proof for the inequality.

Note that there is a difference between real numbers and integers. One should not only prove an inequality for integers (say, by induction) if the variables are real numbers.

1.1 Completing Square

In view of the simple fact $x^2 \geqslant 0$ for all real numbers x, we usually try to group the terms into some squares. Such a way of proving an inequality is called the method of completing square.

Problem 1.1.

(a) Find the minimum value of $x^2 + 2x + 6$ where x is a real number.

(b) Find the minimum value of $3x^2 + x + 1$ where x is a real number.

(c) Find the maximum value of $-4x^2 - 2x + 1$ where x is a real number.

Solution.

(a) The strategy is to write $x^2 + 2x + 6$ as a square plus a constant. Since it is a quadratic polynomial, it can only be the square of a linear polynomial. Hence, we consider
$$x^2 + 2x + 6 = (ax + b)^2 + c$$
for some real constants a, b, c. By expanding and comparing coefficients, both $(x+1)^2+5$ and $(-x-1)^2+5$ are possible choices. They are indeed the same since

$(-y)^2 = y^2$. Then we easily get $x^2 + 2x + 6 = (x+1)^2 + 5 \geqslant 5$, where the minimum is attained when $x = -1$. $\square$

(b) Since we are working on real numbers, the same method applies even if the leading coefficient is not a square (of an integer). Indeed, we can get

$$3x^2 + x + 1 = \left(\sqrt{3}x + \frac{\sqrt{3}}{6}\right)^2 + \frac{11}{12} \geqslant \frac{11}{12},$$

with equality $x = -\frac{1}{6}$. If one finds it difficult to handle radicals, one may deduce the same result by considering

$$3x^2 + x + 1 = 3\left(x^2 + \frac{1}{3}x + \frac{1}{3}\right) = 3\left(x + \frac{1}{6}\right)^2 + \frac{11}{12}.$$

$\square$

(c) If the leading coefficient is positive, then we can find the minimum value. In this case, the leading coefficient is negative. Then we can find the maximum value of the polynomial. The method is just the same. Indeed,

$$-4x^2 - 2x + 1 = -(4x^2 + 2x - 1) = -\left(2x + \frac{1}{2}\right)^2 + \frac{5}{4} \leqslant \frac{5}{4}.$$

Note that $\left(2x + \frac{1}{2}\right)^2 \geqslant 0$ but $-\left(2x + \frac{1}{2}\right)^2 \leqslant 0$, and hence the inequality sign is reversed. Here, the maximum value is attained when $x = -\frac{1}{4}$. $\square$

The method of completing square works perfectly for quadratic polynomials. It fails for polynomials of odd degrees (say, cubic polynomials) since such polynomials have no maximum or minimum values (notice what happens when x tends to either infinity). The following problem shows that it may not work for some polynomials of higher degrees.

Problem 1.2.

(a) Find the minimum value of $x^4 + 6x^3 + 3x^2 - 18x + 2$ where x is a real number.

(b) Prove that $x^4 - 5x^3 + 10x^2 - 12x + 8 \geqslant 0$ for any real number x.

Solution.

(a) We shall use the same way to handle this problem. Since the polynomial is of degree 4, we consider

$$x^4 + 6x^3 + 3x^2 - 18x + 2 = (ax^2 + bx + c)^2 + d$$

for some real constants a, b, c, d. We get 5 equations by comparing coefficients. However, there are only 4 variables and hence there is no solution in general. That means we may not be able to express the polynomial in this way. Luckily, we get a solution in this case. Indeed, we have

$$x^4 + 6x^3 + 3x^2 - 18x + 2 = (x^2 + 3x - 3)^2 - 7 \geqslant -7.$$

Equality holds when $x^2 + 3x - 3 = 0$, i.e. when $x = \dfrac{-3 \pm \sqrt{21}}{2}$. $\square$

(b) If one tries the same method, one will find that there is no solution for a, b, c, d. Indeed, we can only rewrite

$$x^4 - 5x^3 + 10x^2 - 12x + 8 = \left(x^2 - \frac{5}{2}x + \frac{15}{8}\right)^2 - \frac{21}{8}x + \frac{287}{64}.$$

The thing outside the square term does not have a minimum value. We shall use another method to prove this inequality later. $\square$

Sometimes it is more convenient to use the discriminant for help.

Proposition 1.1. Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial where a, b, c are real constants with $a \neq 0$. Then $P(x) \geqslant 0$ for any real number x if any only if $a > 0$ and $\Delta = b^2 - 4ac \leqslant 0$.

This is a simplified version of proposition 5.3 of **The Very Basic Note in Algebra 1.5.1**.

Problem 1.3.

(a) Prove that $20x^2 - 32x + 13 \geqslant 0$ for any real number x.

(b) Find all k such that $(k - 2)x^2 - 2kx + (k + 3) \geqslant 0$ for any real number x.

Solution.

(a) When the coefficients are large, it might be troublesome to solve equations if we try to complete square. Applying proposition 1.1 is a better method. Indeed, the leading coefficient is positive. So we just need to check

$$\Delta = (-32)^2 - 4(20)(13) = -16 < 0.$$

$\square$

(b) Discriminant is also useful when the coefficients are not constants. For this problem, according to proposition 1.1, we need $k - 2 > 0$ (note that $k = 2$ is not possible) and

$$0 \geqslant \Delta = (-2k)^2 - 4(k - 2)(k + 3) = -4k + 24.$$

Clearly, this is the same as $k \geqslant 6$. Note that $k \geqslant 6$ automatically implies $k-2>0$. Hence, any real number $k \geqslant 6$ will make the inequality true. $\square$

Problem 1.4.

(a) Find the minimum value of $3x^2+4y^2+4xy-4x$ where x, y are real numbers.

(b) Prove that $5x^2-2xy+2y^2+2x-4y+2 \geqslant 0$ for any real numbers x, y.

Solution.

(a) The method of completing square applies equally well for more variables, as long as the degree of each variable is 2. Note that only two terms involve y, so it would be simpler to start by considering these terms. So we first consider x as a constant and complete square for $4y^2+4xy$. It is not hard to get $4y^2+4xy=(2y+x)^2-x^2$. Then

$$3x^2+4y^2+4xy-4x=(2y+x)^2+2x^2-4x.$$

Note that the things outside the bracket are independent of y. This is important in order that the bracket can be ignored. For example, if we try to complete square in another way, say, $3x^2+4y^2+4xy-4x=2(x+y)^2+x^2+2y^2-4x$, then minimizing $(x+y)^2$ may not minimize the remaining terms at the same time.

Next, we further work on $2x^2-4x$ to get

$$3x^2+4y^2+4xy-4x=(2y+x)^2+2(x-1)^2-2 \geqslant -2.$$

Equality holds when $2y+x=x-1=0$, i.e. $x=1$ and $y=-\dfrac{1}{2}$. $\square$

(b) There is not much difference whether we work on x or y first. For example, if we view it as a quadratic inequality in x, then we need to prove

$$5x^2+(-2y+2)x+(2y^2-4y+2) \geqslant 0.$$

Since the leading coefficient is positive, it suffices to show $\Delta \leqslant 0$ by proposition 1.1. Indeed,

$$\Delta=(-2y+2)^2-4(5)(2y^2-4y+2)=-36y^2+72y-36=-36(y-1)^2 \leqslant 0$$

by completing square. In general, if there are n variables and we fix one of them (x in this case) as the main variable, then the discriminant Δ will not involve this variable. Hence, we can reduce the original inequality to $\Delta \leqslant 0$, which has one variable less. The same process applies until the inequality is simple enough (or there is a single variable left). $\square$

Problem 1.5.

(a) Prove that $x^2+y^2+z^2 \geqslant xy+yz+zx$ for any real numbers x, y, z.

(b) Prove that $(x+y+z)^2 \geqslant 3(xy+yz+zx)$ for any real numbers x, y, z.

Solution. We always encounter some inequalities where the variables are symmetric to each other and the total degree of each term is the same (such inequality is said to be a homogeneous one). This problem provides some useful facts when proving such inequalities.

(a) We shall provide a standard solution to this problem, i.e.

$$x^2+y^2+z^2-xy-yz-zx = \frac{1}{2}\left((x-y)^2+(y-z)^2+(z-x)^2\right) \geqslant 0.$$

For similar problems, we may use the discriminant method as introduced in the previous problem to deduce the result step by step. Note that equality holds if and only if $x=y=z$. $\square$

(b) Through expansion, the inequality is the same as part (a). Note that this inequality gives a relation between $x+y+z$ and $xy+yz+zx$, which appears frequently in polynomials in 3 variables (these, together with xyz, are called the elementary symmetric polynomials in 3 variables). $\square$

1.2 Factorization

Problem 1.6.

(a) Prove that $x^4-5x^3+10x^2-12x+8 \geqslant 0$ for any real number x.

(b) Prove that $2x^4-x^3-3x^2+x+1 \geqslant 0$ for any positive real number x.

Solution.

(a) This is the problem that we have not yet solved. Apart from completing square, we often try to do factorization in inequality problems. This always helps simplify the expression. For a polynomial of this kind, note that its linear factors can be found by trial and error using the factor theorem. For example, since

$$(2)^4-5(2)^3+10(2)^2-12(2)+8=0,$$

$x-2$ is a factor. Indeed, $x^4-5x^3+10x^2-12x+8=(x-2)(x^3-3x^2+4x-4)$. For inequality problems, the linear factors of a polynomial often (but not always) occur an even number of times. Therefore, it is natural for us to take out another factor $x-2$ to get

$$x^4-5x^3+10x^2-12x+8=(x-2)^2(x^2-x+2).$$

To prove that this is nonnegative, we note that $(x-2)^2 \geqslant 0$ and $x^2 - x + 2 \geqslant 0$, where the second inequality is true by considering $\Delta = -7 < 0$. The product of two nonnegative expressions is nonnegative. Equality holds when $x = 2$ in this case. $\square$

(b) In the same way, we may observe that all of $1, -1, -\dfrac{1}{2}$ are roots of the polynomial. It is not hard to deduce

$$2x^4 - x^3 - 3x^2 + x + 1 = (x-1)^2(x+1)(2x+1) \geqslant 0.$$

Note that the condition $x > 0$ is important for $x+1$ and $2x+1$ to be positive. Equality holds when $x = 1$.

If the coefficients of the polynomial are integers, then its only rational roots can be found using the rational root theorem. In real situation, an inequality of this form usually arises from another inequality for which we can easily guess its equality case. In other words, we usually know what the factors should be before we reduce the problem to this form. $\square$

Problem 1.7. Prove that $x^3 - 2x^2y - 4xy^2 + 8y^3 \geqslant 0$ for any positive reals x, y.

Solution. A homogeneous inequality in 2 variables can be regarded as a polynomial inequality in 1 variable. Indeed, we can use the same way as the previous problem to deduce $x^3 - 2x^2 - 4x + 8 = (x-2)^2(x+2)$. Then we simply need to add the terms y in suitable places (to make the degree of each term equal), i.e.

$$x^3 - 2x^2y - 4xy^2 + 8y^3 = (x-2y)^2(x+2y) \geqslant 0,$$

with equality $x = 2y$. $\square$

Problem 1.8. Let a, b, c be positive real numbers such that $a \geqslant b \geqslant c$. Prove that $a^3b + b^3c + c^3a \geqslant ab^3 + bc^3 + ca^3$.

Solution. We demonstrate how to factorize the polynomial

$$f(a,b,c) = (a^3b + b^3c + c^3a) - (ab^3 + bc^3 + ca^3)$$

in an indirect way. Note that $f(b,b,c) = 0$ by direct computation. Regarding f as a polynomial in a, this means $a = b$ is a root. By the factor theorem, $a - b$ is a factor of f. By symmetry, $b - c$ and $c - a$ are also factors of f. Let

$$f(a,b,c) = (a-b)(b-c)(c-a)g(a,b,c).$$

Since $\deg f = 4$ and $\deg (a-b)(b-c)(c-a) = 3$, we must have $\deg g = 1$, and it is homogeneous. Also, note that both f and $(a-b)(b-c)(c-a)$ are cyclic polynomials (i.e. they remain the same if we replace a, b, c by b, c, a). Thus, g is also cyclic.

The only degree 1 cyclic polynomials in the variables a, b, c are $k(a+b+c)$ for some constant k. Now, by comparing

$$6 = f(2,1,0) = (2-1)(1-0)(0-2)k(2+1+0) = -6k,$$

we obtain $k = -1$. This implies

$$f(a,b,c) = -(a-b)(b-c)(c-a)(a+b+c) \geqslant 0$$

since $a \geqslant b \geqslant c$ and $a+b+c > 0$. Equality holds when two of a, b, c are equal. $\square$

1.3 AM-GM Inequality

Theorem 1.1. (AM-GM inequality(均值不等式)) Let $n \geqslant 2$. For any positive real numbers $a_1, a_2, \ldots, a_n$, we have

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geqslant \sqrt[n]{a_1 a_2 \cdots a_n}.$$

Equality holds if and only if $a_1 = a_2 = \cdots = a_n$.

This is theorem 5.1 of **The Very Basic Note in Algebra 1.5.1**. The AM-GM inequality gives a relationship between the sum and the product of several terms.

Problem 1.9.

(a) Find the minimum value of $x + \dfrac{1}{x}$ where x is a positive real number.

(b) Find the minimum value of $\dfrac{x+y}{z} + \dfrac{y+z}{x} + \dfrac{z+x}{y}$ where x, y, z are positive real numbers.

(c) Find the minimum value of $x + \dfrac{1}{x-2}$ where x is a real number greater than 2.

Solution.

(a) Applying the AM-GM inequality, we can reduce the expression to the product of the terms, which becomes a constant. Indeed, $x + \dfrac{1}{x} \geqslant 2\sqrt{x \cdot \dfrac{1}{x}} = 2$. Equality holds when $x = \dfrac{1}{x}$, i.e. $x = 1$. $\square$

(b) Directly applying the AM-GM inequality to the 3 terms does not give a constant. So we do one more step as follows:

$$\frac{x+y}{z} + \frac{y+z}{x} + \frac{z+x}{y} = \frac{x}{z} + \frac{y}{z} + \frac{y}{x} + \frac{z}{x} + \frac{z}{y} + \frac{x}{y} \geqslant 6\sqrt[6]{\frac{x}{z} \cdot \frac{y}{z} \cdot \frac{y}{x} \cdot \frac{z}{x} \cdot \frac{z}{y} \cdot \frac{x}{y}} = 6.$$

Equality holds when all fractions are equal. One checks that this happens when $x = y = z$. $\square$

(c) Again, the product of the 2 terms is not a constant. To cancel the denominator, we need the term $x - 2$. So we write $x + \frac{1}{x-2} = (x-2) + \frac{1}{x-2} + 2$ and apply the AM-GM inequality to the first 2 terms. This gives

$$x + \frac{1}{x-2} = (x-2) + \frac{1}{x-2} + 2 \geqslant 2\sqrt{(x-2)\cdot\frac{1}{x-2}} + 2 = 4.$$

This minimum value can be attained when $x - 2 = \frac{1}{x-2}$, i.e. when $x = 3$.

Note that we do not apply the AM-GM inequality to all 3 terms simultaneously, i.e.

$$x + \frac{1}{x-2} = (x-2) + \frac{1}{x-2} + 2 \geqslant 3\sqrt[3]{(x-2)\cdot\frac{1}{x-2}\cdot 2} = 3\sqrt[3]{2}.$$

The inequality is still true and we get another lower bound $3\sqrt[3]{2}$ instead of 4. However, this bound is not the minimum value since equality cannot hold. One may check that $x - 2 = \frac{1}{x-2} = 2$ yields no solution. Indeed, $3\sqrt[3]{2} < 4$, and the bound is too weak. $\square$

Problem 1.10. Let x, y, z be positive real numbers satisfying $x + y + z = 6$.

(a) Find the maximum value of xyz.

(b) Find the maximum value of x^3y^2z.

Solution.

(a) In solving inequality problems, we often need to have some relationships between simple expressions, like $x + y + z$ and xyz. The AM-GM inequality provides a simple way for us to do so. Note that we can only find the maximum value of the product when the sum is fixed, but not the minimum (indeed, the product can be sufficiently close to 0). By rewriting the AM-GM inequality, we simply get

$$xyz \leqslant \left(\frac{x+y+z}{3}\right)^3 = 8,$$

with equality $x = y = z$. As $x + y + z = 6$, this can only happen when $x = y = z = 2$. $\square$

(b) If we apply the AM-GM inequality directly, we can only get $x^3y^2z \leqslant \left(\frac{x^3+y^2+z}{3}\right)^3$ where the condition cannot be used. Indeed, to make the powers of x and y equal to 1, we should view x^3 as $x\cdot x\cdot x$, etc. However, there are still problems in the coefficients if we write

$$x^3y^2z = xxxyyz \leqslant \left(\frac{x+x+x+y+y+z}{6}\right)^6.$$

In order to get $x+y+z$ in the numerator, we make one more transformation. In fact, we have

$$x^3y^2z = 108\cdot\frac{x}{3}\cdot\frac{x}{3}\cdot\frac{x}{3}\cdot\frac{y}{2}\cdot\frac{y}{2}\cdot z \leqslant 108\left(\frac{\frac{x}{3}+\frac{x}{3}+\frac{x}{3}+\frac{y}{2}+\frac{y}{2}+z}{6}\right)^6 = 108.$$

Equality holds when $\frac{x}{3}=\frac{y}{2}=z$, i.e. $(x,y,z)=(3,2,1)$ as $x+y+z=6$.

Using similar tricks, we can determine the maximum value for any expression like $x^\alpha y^\beta z^\gamma$. Indeed, it even works if the condition is $ax+by+cz=d$ for some positive a,b,c. $\square$

Problem 1.11. Let x,y,z be positive real numbers satisfying $x^2yz=1$. Find the minimum value of $x+y+2z$.

Solution. This time the case is reversed. We are given the product and try to minimize the sum. But the trick is generally the same. We have

$$x+y+2z = \frac{x}{2}+\frac{x}{2}+y+2z \geqslant 4\sqrt[4]{\frac{x}{2}\cdot\frac{x}{2}\cdot y\cdot 2z} = \frac{4}{\sqrt[4]{2}}.$$

The minimum value $\frac{4}{\sqrt[4]{2}}$ can be attained when $\frac{x}{2}=y=2z$. One checks this happens when $x=\frac{2}{\sqrt[4]{2}}$, $y=\frac{1}{\sqrt[4]{2}}$ and $z=\frac{1}{2\sqrt[4]{2}}$ by using $x^2yz=1$. $\square$

Problem 1.12. Find the maximum value of $x^2(1-x)$ where x is a real number satisfying $0<x<1$.

Solution. If we write $y=1-x>0$, then what we have is $x+y=1$ and we need to maximize x^2y. The same trick in previous problems can be used. Do not think that the problem is always made more complicated by introducing new variables. Certainly, there is no need to do so in this simple problem if one is familiar with such inequalities. By the AM-GM inequality,

$$x^2(1-x) = 4\cdot\frac{x}{2}\cdot\frac{x}{2}\cdot(1-x) \leqslant 4\cdot\left(\frac{\frac{x}{2}+\frac{x}{2}+(1-x)}{3}\right)^3 = \frac{4}{27}.$$

Equality holds when $x=\frac{2}{3}$. Note that $x<1$ is crucial in order that $1-x>0$ (as we need to apply the AM-GM inequality). $\square$

Problem 1.13.

(a) Find the maximum value of $\dfrac{xy^2}{x^2y+y^6+1}$ where x, y are positive real numbers.

(b) Let x, y, z be positive real numbers satisfying $x^2y^2z^2 = x^2+y^2+z^2$. Prove that

$$\frac{x^2+yz}{x^6+5y^3z^3}+\frac{y^2+zx}{y^6+5z^3x^3}+\frac{z^2+xy}{z^6+5x^3y^3} \leqslant \frac{1}{3}.$$

Solution.

(a) The idea is to apply the AM-GM inequality to the terms in the denominator so that something can be cancelled with the numerator. Inspired by the previous problems, we try to write

$$x^2y+y^6+1 = a\cdot\frac{x^2y}{a}+b\cdot\frac{y^6}{b}+c\cdot\frac{1}{c}$$

for some positive integers a, b, c. After applying the AM-GM inequality, we shall get the term $\sqrt[a+b+c]{x^{2a}y^{a+6b}}$ up to a coefficient. In order that it is xy^2, we need $2a = a+b+c$ and $a+6b = 2a+2b+2c$. One way is to take $a=2$, $b=1$, $c=1$. Then we can write the solution neatly as follows:

$$\frac{xy^2}{x^2y+y^6+1} = \frac{xy^2}{\dfrac{x^2y}{2}+\dfrac{x^2y}{2}+y^6+1} \leqslant \frac{xy^2}{4\sqrt[4]{\dfrac{x^2y}{2}\cdot\dfrac{x^2y}{2}\cdot y^6\cdot 1}} = \frac{xy^2}{2\sqrt{2}xy^2} = \frac{\sqrt{2}}{4}.$$

Note that the inequality sign is reversed since the AM-GM inequality is applied in the denominator (recall $x \geqslant y$ implies $\dfrac{1}{x} \leqslant \dfrac{1}{y}$ for $x, y > 0$). Equality holds when $x = \sqrt{2}$ and $y = 1$. $\square$

(b) The same idea applies. Again we shall work on the denominator as the inequality sign is reversed (one may also turn products in numerator into sums). Usually it is not obvious how we should apply the AM-GM inequality and we need some trial and error. Always remember that our goal is to simplify the expressions. As the 3 fractions are the same (up to permutation of the variables), we can concentrate on one of them. Indeed, we observe that

$$x^6+5y^3z^3 = x^6+y^3z^3+y^3z^3+3y^3z^3 \geqslant 3\sqrt[3]{x^6\cdot y^3z^3\cdot y^3z^3}+3y^3z^3 = 3x^2y^2z^2+3y^3z^3$$

by applying the AM-GM inequality to the first 3 terms only. The reason for us to get this expression is that we observe $3x^2y^2z^2+3y^3z^3$ has the factor x^2+yz. Then

$$\frac{x^2+yz}{x^6+5y^3z^3} \leqslant \frac{x^2+yz}{3y^2z^2(x^2+yz)} = \frac{1}{3y^2z^2}.$$

Similarly, the left-hand side of the given inequality is at most

$$\frac{1}{3y^2z^2} + \frac{1}{3z^2x^2} + \frac{1}{3x^2y^2} = \frac{x^2+y^2+z^2}{3x^2y^2z^2} = \frac{1}{3}$$

using the given condition. Equality holds when $x = y = z = \sqrt[4]{3}$. □

1.4 Cauchy-Schwarz Inequality

Theorem 1.2. (Cauchy-Schwarz inequality(柯西不等式)) Let $n \geqslant 2$. For any real numbers $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$, we have

$$(a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2) \geqslant (a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2.$$

Equality holds if and only if $a_ib_j = a_jb_i$ for any $i \neq j$.

This is theorem 5.2 of **The Very Basic Note in Algebra 1.5.1**. The Cauchy-Schwarz inequality has far more applications since it can be applied to any real numbers instead of only nonnegative real numbers.

Problem 1.14.

(a) Prove that $x_1^2 + x_2^2 + \cdots + x_n^2 \geqslant \frac{(x_1 + x_2 + \cdots + x_n)^2}{n}$ for any real numbers $x_1, x_2, \ldots, x_n$.

(b) Prove that $x_1^3 + x_2^3 + \cdots + x_n^3 \geqslant \frac{(x_1 + x_2 + \cdots + x_n)^3}{n^2}$ for any positive real numbers $x_1, x_2, \ldots, x_n$.

Solution.

(a) In view of the Cauchy-Schwarz inequality, each term on the right-hand side (the smaller side) is the square root of the product of the 2 corresponding terms on the left-hand side. For this problem, we try to make x_j to be the square root of the product of 2 terms, one of which is x_j^2. Therefore, we may consider the other term to be the constant 1. In other words, we consider

$$(1^2 + 1^2 + \cdots + 1^2)(x_1^2 + x_2^2 + \cdots + x_n^2) \geqslant (x_1 + x_2 + \cdots + x_n)^2$$

by the Cauchy-Schwarz inequality. As $1^2 + 1^2 + \cdots + 1^2 = n$, the inequality is the one we want, with equality case $1 : 1 : \cdots : 1 = x_1 : x_2 : \cdots : x_n$, i.e. $x_1 = x_2 = \cdots = x_n$.

This inequality gives the relationship between the sum of squares and sum of n terms, which is quite useful. We may also rewrite it as

$$\sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}} \geqslant \frac{x_1 + x_2 + \cdots + x_n}{n},$$

which asserts the quadratic mean is no less than the arithmetic mean. □

(b) If we use the same consideration as above, we may consider

$$\left(\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}\right)(x_1^3+x_2^3+\cdots+x_n^3) \geqslant (x_1+x_2+\cdots+x_n)^2.$$

That means the two sequences are $\frac{1}{\sqrt{x_1}}, \frac{1}{\sqrt{x_2}}, \ldots, \frac{1}{\sqrt{x_n}}$ and $\sqrt{x_1}^3, \sqrt{x_2}^3, \ldots, \sqrt{x_n}^3$ (note that the variables need to be positive). However, the term $\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}$ does not make things simpler. We cannot use $1, 1, \ldots, 1$ as above, otherwise we get something like $\sqrt{x_j}^3$. Indeed, we apply the Cauchy-Schwarz inequality as follows:

$$(x_1+x_2+\cdots+x_n)(x_1^3+x_2^3+\cdots+x_n^3) \geqslant (x_1^2+x_2^2+\cdots+x_n^2)^2.$$

The reason is that the new thing $x_1^2+x_2^2+\cdots+x_n^2$ arisen has lower degree than the original one, which seems to be simpler. This can be further simplified using part (a). This gives

$$x_1^3+x_2^3+\cdots+x_n^3 \geqslant \frac{(x_1^2+x_2^2+\cdots+x_n^2)^2}{x_1+x_2+\cdots+x_n} \geqslant \frac{(x_1+x_2+\cdots+x_n)^4}{n^2(x_1+x_2+\cdots+x_n)} = \frac{(x_1+x_2+\cdots+x_n)^3}{n^2}.$$

For equality, we need $x_1 = x_2 = \cdots = x_n$ as we have used part (a). □

Problem 1.15.

(a) Let $x_1, x_2, \ldots, x_n$ be positive real numbers satisfying $x_1+x_2+\cdots+x_n = n$. Prove that $\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n} \geqslant n$.

(b) Let $x_1, x_2, \ldots, x_n$ be positive real numbers satisfying $x_1+x_2+\cdots+x_n = 1$. Prove that $\frac{x_1^2}{x_2+1}+\frac{x_2^2}{x_3+1}+\cdots+\frac{x_{n-1}^2}{x_n+1}+\frac{x_n^2}{x_1+1} \geqslant \frac{1}{n+1}$.

(c) Let x, y, z be positive real numbers satisfying $x+y+z = 3$. Prove that

$$\frac{x}{y+1}+\frac{y}{z+1}+\frac{z}{x+1} \geqslant \frac{3}{2}.$$

Solution. The Cauchy-Schwarz inequality is useful in clearing denominators. This problem provides some examples.

(a) We may consider the left-hand side of the inequality as one bracket in the Cauchy-Schwarz inequality. For the other bracket, we suppose each term is just the denominator of the corresponding term, so that the products of corresponding terms are 1. In other words, we get

$$(x_1+x_2+\cdots+x_n)\left(\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}\right) \geqslant (1+1+\cdots+1)^2 = n^2.$$

Using the condition, this gives $\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} \geqslant n$ as desired. Equality holds when $x_1 = x_2 = \cdots = x_n = 1$. $\square$

(b) For the same reason, we consider

$$\big((x_2+1)+(x_3+1)+\cdots+(x_1+1)\big)\left(\frac{x_1^2}{x_2+1}+\frac{x_2^2}{x_3+1}+\cdots+\frac{x_n^2}{x_1+1}\right)$$
$$\geqslant (x_1+x_2+\cdots+x_n)^2 = 1.$$

Note that the first bracket is just $x_1+x_2+\cdots+x_n+n = n+1$. The inequality then follows, with equality $x_1 = x_2 = \cdots = x_n = \frac{1}{n}$. $\square$

(c) If we try the same method, we will get

$$\big((y+1)+(z+1)+(x+1)\big)\left(\frac{x}{y+1}+\frac{y}{z+1}+\frac{z}{x+1}\right) \geqslant (\sqrt{x}+\sqrt{y}+\sqrt{z})^2.$$

This may not be a good choice due to the radicals arisen. To avoid such terms, we try to multiply something to the denominator to make up the first bracket. The goal is to make the right-hand side simpler. For example, we may have

$$\big(x(y+1)+y(z+1)+z(x+1)\big)\left(\frac{x}{y+1}+\frac{y}{z+1}+\frac{z}{x+1}\right) \geqslant (x+y+z)^2 = 9.$$

That means we have $\frac{x}{y+1}+\frac{y}{z+1}+\frac{z}{x+1} \geqslant \frac{9}{xy+yz+zx+3}$. In order to get the lower bound, we need to find an upper bound for $xy+yz+zx$. As we have $x+y+z=3$, we may apply problem 1.5(b) to get $xy+yz+zx \leqslant \frac{1}{3}(x+y+z)^2 = 3$. This yields $\frac{x}{y+1}+\frac{y}{z+1}+\frac{z}{x+1} \geqslant \frac{9}{3+3} = \frac{3}{2}$. Equality holds when $x=y=z=1$. $\square$

Problem 1.16. (Nesbitt's inequality(内斯比特不等式)) Let a, b, c be positive real numbers. Prove that

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} \geqslant \frac{3}{2}.$$

Solution. We do a bit of transformations before applying the Cauchy-Schwarz inequality. Indeed, note that

$$\begin{aligned}
& \frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} \geqslant \frac{3}{2} \\
\Leftrightarrow \quad & \frac{a+b+c}{b+c}+\frac{b+c+a}{c+a}+\frac{c+a+b}{a+b} \geqslant \frac{9}{2} \\
\Leftrightarrow \quad & (a+b+c)\left(\frac{1}{b+c}+\frac{1}{c+a}+\frac{1}{a+b}\right) \geqslant \frac{9}{2} \\
\Leftrightarrow \quad & \big((b+c)+(c+a)+(a+b)\big)\left(\frac{1}{b+c}+\frac{1}{c+a}+\frac{1}{a+b}\right) \geqslant 9.
\end{aligned}$$

This holds by the Cauchy-Schwarz inequality. Equality holds when $a = b = c$. $\square$

Next, the Cauchy-Schwarz inequality can also be used to clear square roots. In view of problem 1.14(a), if we replace each x_j by $\sqrt{x_j}$ for $x_i > 0$ and rewrite the inequality, we get

$$\sqrt{x_1} + \sqrt{x_2} + \cdots + \sqrt{x_n} \leqslant \sqrt{n(x_1 + x_2 + \cdots + x_n)}.$$

Problem 1.17. Let x, y, z be positive real numbers satisfying $x + y + z = 3$. Prove that

$$\sqrt{xy+1} + \sqrt{yz+1} + \sqrt{zx+1} \leqslant 3\sqrt{2}.$$

Solution. Using the Cauchy-Schwarz inequality as explained, we have

$$\sqrt{xy+1} + \sqrt{yz+1} + \sqrt{zx+1} \leqslant \sqrt{3(xy+1+yz+1+zx+1)}.$$

Therefore, it suffices to prove $3(xy + yz + zx + 3) \leqslant (3\sqrt{2})^2$, i.e. $xy + yz + zx \leqslant 3$. This is true since $xy + yz + zx \leqslant \frac{1}{3}(x + y + z)^2 = 3$. $\square$

Problem 1.18. Prove that $\sqrt{2(x+y)} + \sqrt{3(x+2y)} \geqslant 2\sqrt{x} + 3\sqrt{y}$ for any positive real numbers x, y.

Solution. In view of the inequality sign, we may start from the right-hand side. Then we get $\sqrt{x} + \sqrt{y} \leqslant \sqrt{2(x+y)}$ as above. So one of the terms on the left-hand side can be eliminated and it suffices to prove $\sqrt{3(x+2y)} \geqslant \sqrt{x}+2\sqrt{y}$. This is more or less similar to the inequality we have just used. Therefore, we may try to generalize the idea to clear radicals. Recall that the inequality $\sqrt{2(x+y)} \geqslant \sqrt{x} + \sqrt{y}$ comes from

$$(1+1)(x+y) \geqslant (\sqrt{x} + \sqrt{y})^2$$

by the Cauchy-Schwarz inequality. In order to get a similar inequality, we may try

$$(1+2)(x+2y) \geqslant (\sqrt{x} + 2\sqrt{y})^2.$$

This is just the inequality we are looking for. By following the inequalities used, we find that equality holds when $x = y$. $\square$

1.5 Homogenization

Homogenization refers to the process of making the expressions homogeneous, i.e. making every term have the same degree. This can usually be done by using the conditions (and cannot be done if there is no additional information). The reason for this process is that we observe both the AM-GM inequality and the Cauchy-Schwarz inequality (and many other famous inequalities) are homogeneous in the variables.

We first go through some ways to handle homogeneous inequalities (mainly in 3 variables). The best way to handle symmetric inequalities in 3 variables is to use Muirhead's theorem and Schur's inequality (see **Inequalities 5.0**), but we are not going to introduce these results in this note. Here, we only demonstrate how to use the AM-GM inequality.

Problem 1.19. Let a, b, c be positive real numbers.

(a) Prove that $a^3 + b^3 + c^3 \geqslant 3abc$.

(b) Prove that $2(a^3 + b^3 + c^3) \geqslant a^2b + a^2c + b^2a + b^2c + c^2a + c^2b$.

(c) Prove that $a^4b + a^4c + b^4a + b^4c + c^4a + c^4b \geqslant 2(a^2b^2c + a^2bc^2 + ab^2c^2)$.

Solution.

(a) This is obvious by applying the AM-GM inequality directly as follows:

$$a^3 + b^3 + c^3 \geqslant 3\sqrt[3]{a^3b^3c^3} = 3abc.$$

□

(b) Due to symmetry, we just need to consider how to make up a term a^2b from a^3 and b^3 (it is likely that c^3 is not needed). As the ratio of powers of a and b should be $2:1$, we use the AM-GM inequality to obtain $a^3+a^3+b^3 \geqslant 3a^2b$. This suggests how other terms can be made. We have

$$\begin{aligned}
a^3 + a^3 + b^3 &\geqslant 3a^2b,\\
a^3 + a^3 + c^3 &\geqslant 3a^2c,\\
b^3 + b^3 + a^3 &\geqslant 3b^2a,\\
b^3 + b^3 + c^3 &\geqslant 3b^2c,\\
c^3 + c^3 + a^3 &\geqslant 3c^2a,\\
c^3 + c^3 + b^3 &\geqslant 3c^2b.
\end{aligned}$$

Adding these inequalities and dividing both sides by 3, we get the inequality as desired. Equality holds when $a = b = c$. In general, for a homogeneous expression of degree n, $a^n+b^n+c^n$ is the 'strongest' one in the sense that it is no less than other symmetric expression of degree n up to a coefficient. The same consideration in this solution applies equally well for higher degrees. □

(c) Again, we try to make up a^2b^2c from the terms on the left-hand side. To avoid guessing, we may consider a more general form

$$\alpha(a^4b + ab^4) + \beta(a^4c + b^4c) + \gamma(ac^4 + bc^4)$$

for some integers $\alpha, \beta, \gamma \geqslant 0$. Note that a^4b, ab^4 should have the same coefficient, etc., since a, b have the same power in a^2b^2c. Applying the AM-GM inequality to these $2(\alpha+\beta+\gamma)$ terms, this is at least

$$\sqrt[2(\alpha+\beta+\gamma)]{a^{4\alpha}b^{\alpha} \cdot a^{\alpha}b^{4\alpha} \cdot a^{4\beta}c^{\beta} \cdot b^{4\beta}c^{\beta} \cdot a^{\gamma}c^{4\gamma} \cdot b^{\gamma}c^{4\gamma}} = a^{\frac{5\alpha+4\beta+\gamma}{2(\alpha+\beta+\gamma)}} b^{\frac{5\alpha+4\beta+\gamma}{2(\alpha+\beta+\gamma)}} c^{\frac{2\beta+8\gamma}{2(\alpha+\beta+\gamma)}}.$$

In order to get a^2b^2c, we need $5\alpha+4\beta+\gamma = 4\alpha+4\beta+4\gamma$ and $2\beta+8\gamma = 2\alpha+2\beta+2\gamma$. For example, we can take $\alpha = 3$, $\beta = 0$ and $\gamma = 1$. Due to symmetry, we obtain

$$\begin{aligned} 3a^4b + 3ab^4 + ac^4 + bc^4 &\geqslant 8a^2b^2c, \\ 3b^4c + 3bc^4 + ba^4 + ca^4 &\geqslant 8ab^2c^2, \\ 3c^4a + 3ca^4 + cb^4 + ab^4 &\geqslant 8a^2bc^2. \end{aligned}$$

Adding these and dividing both sides by 4, we get the desired inequality, with equality $a = b = c$. □

The above trick can be used even if the inequality is cyclic but not symmetric.

Problem 1.20. Let a, b, c be positive real numbers satisfying $a + b + c = 3$. Prove that

$$2(a^3 + b^3 + c^3) \geqslant a^2 + b^2 + c^2 + 3.$$

Solution. First, we homogenize the inequality using $a+b+c = 3$. The highest degree is 3. To make $a^2 + b^2 + c^2$ have degree 3, we simply multiply $\frac{a+b+c}{3} = 1$ to it. Similarly, we multiply the constant 3 by $\frac{(a+b+c)^3}{27}$. That means we shall prove

$$2(a^3 + b^3 + c^3) \geqslant \frac{(a^2+b^2+c^2)(a+b+c)}{3} + \frac{(a+b+c)^3}{9}.$$

After expansion and simplification, this is the same as

$$7(a^3 + b^3 + c^3) \geqslant 3(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b) + 3abc.$$

Note that since the inequality is symmetric, the coefficients of a^2b, a^2c, etc., should be the same. This means to expand everything, it suffices to find the coefficients of a^3, a^2b and abc.

Now, we make use of the results of problem 1.19 to get

$$\begin{aligned} 6(a^3 + b^3 + c^3) &\geqslant 3(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b), \\ a^3 + b^3 + c^3 &\geqslant 3abc. \end{aligned}$$

This proves the inequality by adding them. Equality holds when $a = b = c = 1$.

Note that in the solution, the condition $a + b + c = 3$ can be ignored after homogenization. This is because we can always multiply each variable by the same constant to get $a+b+c = 3$ while the inequality is not affected. □

Problem 1.21. (CWMO 2001 Problem 4) Let x, y, z be positive real numbers such that $x + y + z \geqslant xyz$. Find the minimum value of $\dfrac{x^2 + y^2 + z^2}{xyz}$.

Solution. The expression has degree -1, while the difference of degrees on both sides of the condition is $1 - 3 = -2$. To carry out homogenization, we consider the square of the expression instead (so that it has degree -2). Then we can use the condition $1 \geqslant \dfrac{xyz}{x + y + z}$ to homogenize it. Indeed, we have

$$\left(\frac{x^2 + y^2 + z^2}{xyz}\right)^2 \geqslant \frac{(x^2 + y^2 + z^2)^2}{x^2y^2z^2} \cdot \frac{xyz}{x + y + z} = \frac{(x^2 + y^2 + z^2)^2}{xyz(x + y + z)}.$$

We guess the minimum holds when $x = y = z$ since this is the most common equality case. So we guess $\dfrac{(x^2 + y^2 + z^2)^2}{xyz(x + y + z)} \geqslant 3$, which means we should prove

$$x^4 + y^4 + z^4 + 2(x^2y^2 + y^2z^2 + z^2x^2) \geqslant 3(x^2yz + xy^2z + xyz^2).$$

The same trick used in problem 1.19 can be applied. We have

$$\begin{aligned} x^4 + x^4 + y^4 + z^4 &\geqslant 4\sqrt[4]{(x^4)(x^4)(y^4)(z^4)} = 4x^2yz, \\ x^2y^2 + x^2z^2 &\geqslant 2\sqrt{(x^2y^2)(x^2z^2)} = 2x^2yz. \end{aligned}$$

Adding similar inequalities, we get

$$\begin{aligned} x^4 + y^4 + z^4 &\geqslant x^2yz + xy^2z + xyz^2, \\ 2(x^2y^2 + y^2z^2 + z^2x^2) &\geqslant 2(x^2yz + xy^2z + xyz^2). \end{aligned}$$

This proves the desired inequality. Thus, the minimum value of $\dfrac{x^2 + y^2 + z^2}{xyz}$ is $\sqrt{3}$, with equality $x = y = z = \sqrt{3}$. □

2 Sequences

There may be some problems which define some sequences and ask us to prove some properties of these sequences. Although it may not be always possible for us to find the general term, we should do so whenever we can.

Problem 2.1.

(a) Let $a_1 = 2$ and $a_{n+1} = 2a_n + 1$ for $n \geqslant 1$. Find a_n.

(b) (CSMO 2008 Problem 2) Suppose the sequence $\{a_n\}$ satisfies

$$a_1 = 1,\ a_{n+1} = 2a_n + n \cdot (1 + 2^n),\ n = 1, 2, 3, \ldots.$$

Find the general term a_n.

Solution.

(a) Firstly, we have $a_n = 2a_{n-1} + 1$. Next, we express a_{n-1} by using $a_{n-1} = 2a_{n-2} + 1$. In this way, we can eventually express a_n as a_1, which is a known constant. In order to do this systematically, we consider the following equations:

$$\begin{aligned} a_n &= 2a_{n-1} + 1, \\ 2a_{n-1} &= 4a_{n-2} + 2, \\ 4a_{n-2} &= 8a_{n-3} + 4, \\ &\ldots, \\ 2^{n-2}a_2 &= 2^{n-1}a_1 + 2^{n-2}. \end{aligned}$$

When these equations are added, all intermediate terms are cancelled (by the choice of the coefficients). Then we get

$$a_n = 2^{n-1}a_1 + (1 + 2 + 4 + \cdots + 2^{n-2}) = 2^n + 2^{n-1} - 1.$$

Here, we have used the sum of geometric sequences

$$a + ar + ar^2 + \cdots + ar^k = a \cdot \frac{r^{k+1} - 1}{r - 1}$$

for $r \neq 1$. □

(b) In the same way, we have

$$\begin{aligned} a_n &= 2a_{n-1} + (n-1) \cdot (1 + 2^{n-1}), \\ 2a_{n-1} &= 4a_{n-2} + (n-2) \cdot (2 + 2^{n-1}), \\ 4a_{n-2} &= 8a_{n-3} + (n-3) \cdot (4 + 2^{n-1}), \\ &\ldots, \\ 2^{n-2}a_2 &= 2^{n-1}a_1 + (2^{n-2} + 2^{n-1}). \end{aligned}$$

Then we get

$$\begin{aligned}a_n &= 2^{n-1}a_1 + (n-1)\cdot(1+2^{n-1}) + (n-2)\cdot(2+2^{n-1}) + \cdots + (2^{n-2}+2^{n-1})\\ &= 2^{n-1} + \sum_{k=1}^{n-1} k(2^{n-1-k}+2^{n-1}).\end{aligned}$$

It remains to compute the summation. Firstly, we have

$$\begin{aligned}\sum_{k=1}^{n-1} k\cdot 2^{n-1-k} &= (1) + (1+2) + (1+2+2^2) + \cdots + (1+2+2^2+\cdots+2^{n-2})\\ &= (2-1) + (2^2-1) + (2^3-1) + \cdots + (2^{n-1}-1)\\ &= (2+2^2+2^3+\cdots+2^{n-1}) - (n-1)\\ &= 2^n - n - 1.\end{aligned}$$

Secondly, it is clear that

$$\sum_{k=1}^{n-1} k\cdot 2^{n-1} = 2^{n-1}\cdot\frac{n(n-1)}{2} = n(n-1)2^{n-2}.$$

Thus, we obtain

$$a_n = 2^{n-1} + (2^n - n - 1) + n(n-1)2^{n-2} = (n^2 - n + 6)\cdot 2^{n-2} - n - 1.$$

□

Problem 2.2.

(a) Let $a_1 = 1$ and $a_{n+1} = a_1 + a_2 + \cdots + a_n + 2$ for $n \geqslant 1$. Find a_n for $n \geqslant 2$.

(b) (CWMO 2004 Problem 5) Suppose the sequence $\{a_n\}$ satisfies the conditions $a_1 = a_2 = 1$ and

$$a_{n+2} = \frac{1}{a_{n+1}} + a_n,\ n = 1, 2, \ldots.$$

Find a_{2004}.

Solution. Sometimes we can guess the formula and prove it by induction. Here are two examples.

(a) We compute a few terms as follows: $\{a_n\} = \{1, 3, 6, 12, 24, \ldots\}$. Thus, we guess the general formula $a_n = 3\cdot 2^{n-2}$ for $n \geqslant 2$. The base case $n = 2$ holds since $a_2 = 3$. Assume $a_n = 3\cdot 2^{n-2}$ for $2 \leqslant n \leqslant k$, where $k \geqslant 2$ is fixed. Then we have

$$\begin{aligned}a_{k+1} &= a_1 + a_2 + \cdots + a_k + 2\\ &= 1 + 3 + 3\cdot 2 + 3\cdot 2^2 + \cdots + 3\cdot 2^{k-2} + 2\\ &= 1 + 3(2^{k-1}-1) + 2\\ &= 3\cdot 2^{k-1}.\end{aligned}$$

So the formula is true for the case $n = k+1$. Inductively, the formula is true for any $n \geqslant 2$. □

(b) We compute some small terms $\{a_n\} = \left\{1, 1, 2, \frac{3}{2}, \frac{8}{3}, \frac{15}{8}, \frac{48}{15}, \frac{105}{48}, \ldots\right\}$. With some more observations, we may guess how the numbers are formed. For example, we observe $8 = 2 \times 4$, $48 = 2 \times 4 \times 6$, $15 = 3 \times 5$, $105 = 3 \times 5 \times 7$, etc. If we denote $(2k+1)!! = 1 \times 3 \times 5 \times \cdots \times (2k+1)$ and $(2k)!! = 2 \times 4 \times 6 \times \cdots \times (2k)$, then we guess $a_n = \frac{(n-1)!!}{(n-2)!!}$ for $n \geqslant 2$. The base cases $n = 2, 3$ clearly hold. For the inductive step, assume $a_k = \frac{(k-1)!!}{(k-2)!!}$ and $a_{k+1} = \frac{k!!}{(k-1)!!}$. Then we have

$$a_{k+2} = \frac{1}{a_{k+1}} + a_k = \frac{(k-1)!!}{k!!} + \frac{(k-1)!!}{(k-2)!!} = \frac{(k-1)!!(1+k)}{k!!} = \frac{(k+1)!!}{k!!}.$$

This proves the formula. In particular, $a_{2004} = \frac{2003!!}{2002!!}$. □

Problem 2.3. (JBMO Shortlist 2020 A2) Consider the sequence $a_1, a_2, a_3, \ldots$ defined by $a_1 = 9$ and

$$a_{n+1} = \frac{(n+5)a_n + 22}{n+3}$$

for $n \geqslant 1$. Find all natural numbers n for which a_n is a perfect square of an integer.

Solution. Once again, the first step is to guess a formula for a_n. The first few values of the sequence are $9, 19, 31, 45, 61, \ldots$. The differences of consecutive terms are $19 - 9 = 10$, $31 - 19 = 12$, $45 - 31 = 14$, $61 - 45 = 16$, etc. Therefore, we guess

$$a_n = 9 + \Big(10 + 12 + \cdots + \big(10 + 2(n-2)\big)\Big) = 9 + \frac{\big(10 + (2n+6)\big)(n-1)}{2} = n^2 + 7n + 1.$$

We can prove this by induction. The base case $n = 1$ clearly holds. Assuming $a_k = k^2 + 7k + 1$, we have

$$\begin{aligned} a_{k+1} &= \frac{(k+5)a_k + 22}{k+3} = \frac{(k+5)(k^2+7k+1) + 22}{k+3} \\ &= \frac{k^3 + 12k^2 + 36k + 27}{k+3} = k^2 + 9k + 9 = (k+1)^2 + 7(k+1) + 1. \end{aligned}$$

Therefore, the formula holds for any n by induction.

It remains to find all n such that $n^2 + 7n + 1$ is a perfect square. One standard way is to bound this between two perfect squares, so that there is only a finite number of possibilities. Indeed, we find that

$$(n+1)^2 = n^2 + 2n + 1 < n^2 + 7n + 1 < n^2 + 8n + 16 = (n+4)^2.$$

Therefore, $n^2 + 7n + 1$ can only be $(n+2)^2$ or $(n+3)^2$. Thus, we obtain $n = 1, 8$. Correspondingly, $n^2 + 7n + 1 = 3^2, 11^2$. So the answers are $n = 1, 8$.

As can be seen, a harder problem is usually just a combination of several elementary problems. We just need to find the correct way to solve the problem. □

Problem 2.4. (CSMO 2011 Problem 7) Suppose the sequence $\{a_n\}$ satisfies

$$a_1 = a_2 = 1,\ a_n = 7a_{n-1} - a_{n-2},\ n \geqslant 3.$$

Prove that for each $n \in \mathbb{N}$, $a_n + a_{n+1} + 2$ is a perfect square.

Solution. We have $\{a_n\} = \{1, 1, 6, 41, 281, 1926, \dots\}$. Consider $b_n^2 = a_n + a_{n+1} + 2$. Then $\{b_n\} = \{2, 3, 7, 18, 47, \dots\}$. The new sequence $\{b_n\}$ may have similar form as the original sequence. Hence, it is not hard to observe $b_n = 3b_{n-1} - b_{n-2}$ for $n \geqslant 3$.

Therefore, we define $\{b_n\}$ by $b_1 = 2$, $b_2 = 3$ and $b_n = 3b_{n-1} - b_{n-2}$ for $n \geqslant 3$. We shall prove

$$b_n^2 = a_n + a_{n+1} + 2.$$

There may be simpler ways to prove this fact. But it is more natural to use the general formulae since all these can be found. Recall how we can find the general term of a linear recurrence sequence. For $\{a_n\}$, the roots of the characteristic equation $x^2 - 7x + 1 = 0$ are $\dfrac{7 \pm 3\sqrt{5}}{2}$. This implies

$$a_n = A\left(\frac{7+3\sqrt{5}}{2}\right)^n + B\left(\frac{7-3\sqrt{5}}{2}\right)^n$$

for some constants A, B. Using the values for $n = 1, 2$, we can solve the equations to get

$$a_n = \frac{9-4\sqrt{5}}{3}\left(\frac{7+3\sqrt{5}}{2}\right)^n + \frac{9+4\sqrt{5}}{3}\left(\frac{7-3\sqrt{5}}{2}\right)^n.$$

Similarly, we obtain

$$b_n = \frac{3-\sqrt{5}}{2}\left(\frac{3+\sqrt{5}}{2}\right)^n + \frac{3+\sqrt{5}}{2}\left(\frac{3-\sqrt{5}}{2}\right)^n.$$

We then check that

$$b_n^2 = \frac{7-3\sqrt{5}}{2}\left(\frac{7+3\sqrt{5}}{2}\right)^n + \frac{7+3\sqrt{5}}{2}\left(\frac{7-3\sqrt{5}}{2}\right)^n + 2$$

and

$$\begin{aligned}
&a_n + a_{n+1} + 2 \\
&= \left(\frac{9-4\sqrt{5}}{3} + \frac{3-\sqrt{5}}{6}\right)\left(\frac{7+3\sqrt{5}}{2}\right)^n + \left(\frac{9+4\sqrt{5}}{3} + \frac{3+\sqrt{5}}{6}\right)\left(\frac{7-3\sqrt{5}}{2}\right)^n + 2 \\
&= \frac{7-3\sqrt{5}}{2}\left(\frac{7+3\sqrt{5}}{2}\right)^n + \frac{7+3\sqrt{5}}{2}\left(\frac{7-3\sqrt{5}}{2}\right)^n + 2.
\end{aligned}$$

The two quantities are equal and hence $a_n + a_{n+1} + 2$ is a square (by noting that all b_n's are integers). $\square$

Problem 2.5. Define the sequence $\{a_n\}$ by $a_1 = 1$ and

$$a_{n+1} = \frac{a_n^2 + 2}{2a_n}$$

for $n \geqslant 1$. Prove that $\{a_n\}$ is strictly decreasing for $n \geqslant 2$.

Solution. It is not possible to find a general term of the sequence. Clearly, all terms are positive. To prove that $a_n > a_{n+1}$, we note that

$$\begin{aligned} & a_n > a_{n+1} \\ \Leftrightarrow \qquad & a_n > \frac{a_n^2 + 2}{2a_n} \\ \Leftrightarrow \qquad & 2a_n^2 > a_n^2 + 2 \\ \Leftrightarrow \qquad & a_n > \sqrt{2}. \end{aligned}$$

Indeed, we can prove by induction that $a_n > \sqrt{2}$ for all $n \geqslant 2$. The base case is $a_2 = \dfrac{3}{2} > \sqrt{2}$. Assuming $a_k > \sqrt{2}$, we have

$$a_{k+1} - \sqrt{2} = \frac{a_k^2 + 2}{2a_k} - \sqrt{2} = \frac{a_k^2 - 2\sqrt{2}a_k + 2}{2a_k} = \frac{(a_k - \sqrt{2})^2}{2a_k} \geqslant 0.$$

Note that equality cannot hold since $a_k > \sqrt{2}$ by the inductive hypothesis. This proves $a_n > \sqrt{2}$ for all n, and hence $a_n > a_{n+1}$ for all n. $\square$

3 Functional Equations

Sometimes we may be asked to find all functions which satisfy certain conditions (could be some equations or inequalities that are true for any x, y, etc.). That means we need to determine $f(x)$ for any x in the specified domain. The most usual way to solve such problems is to substitute various values of x, y until we get enough information to deduce the result.

Unless some additional conditions are given (which rarely happens), we cannot assume the function is 'nice'. For example, we cannot assume it is a polynomial, invertible, or continuous, etc. However, we can normally guess the answers by considering some nice solutions first.

Problem 3.1.

(a) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+1) = x^2 - 1$ for all $x \in \mathbb{R}$.

(b) Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f\left(\frac{1}{x}\right) = 2x + 3$ for all $x \in \mathbb{R}^+$.

Solution.

(a) Let $y = x+1$. Then $x = y-1$ so that $f(y) = (y-1)^2 - 1 = y^2 - 2y$ for any $y \in \mathbb{R}$. $\square$

(b) Let $y = \frac{1}{x}$. Then $x = \frac{1}{y}$ so that $f(y) = \frac{2}{y} + 3$ for any $y \in \mathbb{R}^+$. $\square$

Note that we have to be careful about the domain of the variables. For example, if we are given $f(x^2) = x^2 + 1$ for any $x \in \mathbb{R}$, then we use the same way and let $y = x^2$ to deduce $f(y) = y + 1$. However, here we only have $f(y) = y + 1$ for any $y \geqslant 0$. But there is no information for $f(y)$ where $y < 0$. So we should always take care of the criteria of the variables at each step.

Problem 3.2. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x) + 2f\left(\frac{1}{x}\right) = 2x + \frac{1}{x}$ for all $x \in \mathbb{R}^+$.

Solution. It is given that

$$f(x) + 2f\left(\frac{1}{x}\right) = 2x + \frac{1}{x}. \tag{1}$$

Replace x by $\frac{1}{x}$ in (1). We get

$$f\left(\frac{1}{x}\right) + 2f(x) = \frac{2}{x} + x. \tag{2}$$

We can view $f(x)$ and $f\left(\frac{1}{x}\right)$ as variables in equations (1) and (2). Indeed, we compute $2 \times (2) - (1)$ to obtain

$$3f(x) = 2\left(\frac{2}{x} + x\right) - \left(2x + \frac{1}{x}\right) = \frac{3}{x}.$$

This gives $f(x) = \frac{1}{x}$ for any $x \in \mathbb{R}^+$. Note that we use the conditions to deduce that this is a possible solution. However, there could possibly be no solution at all. Therefore, we always need to check if this is a solution. Indeed, when $f(x) = \frac{1}{x}$,

$$f(x) + 2f\left(\frac{1}{x}\right) = \frac{1}{x} + 2x$$

so that the condition is satisfied. $\square$

Problem 3.3. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y + f(y)) = 2x + f(f(y))$$

for all $x, y \in \mathbb{R}$.

Solution. Normally we start by putting some simple values. For example, by putting $y = 0$, we obtain

$$f(x + f(0)) = 2x + f(f(0)).$$

This is basically the same as problem 3.1. Replacing x by $x - f(0)$, we get

$$f(x) = 2x - 2f(0) + f(f(0)).$$

This shows $f(x) = 2x + c$ for some constant c. Now, for any such function, we check that

$$\begin{aligned} f(x + y + f(y)) &= 2(x + y + 2y + c) + c = 2x + 6y + 3c, \\ 2x + f(f(y)) &= 2x + 2(2y + c) + c = 2x + 4y + 3c. \end{aligned}$$

The two sides are not equal for $y \neq 0$. Thus, there is no such function. This shows the importance of checking solutions. $\square$

Problem 3.4. (Cauchy equation(柯西方程)) Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that

$$f(x + y) = f(x) + f(y)$$

for all $x, y \in \mathbb{Q}$.

Solution. It is given that

$$f(x+y) = f(x) + f(y). \tag{1}$$

By putting $y = 0$ in (1), we have $f(x) = f(x) + f(0)$, i.e. $f(0) = 0$. Putting $y = -x$ in (1) and using $f(0) = 0$, we get $f(-x) = -f(x)$.

Next, define $c = f(1)$. We shall prove by induction that $f(k) = ck$ for $k \in \mathbb{Z}^+$. The base case $f(1) = c$ holds trivially. Assuming $f(k) = ck$, we have

$$f(k+1) = f(k) + f(1) = ck + c = c(k+1).$$

This proves $f(k) = ck$ for all $k \in \mathbb{Z}^+$. Also, since $f(-x) = -f(x)$ and $f(0) = 0$, this is indeed true for all $k \in \mathbb{Z}$.

For the case of $\mathbb{Q}$, we take any $\frac{m}{n} \in \mathbb{Q}^+$ with $m, n \in \mathbb{Z}^+$. In the same way as above, we can prove $f\left(\frac{k}{n}\right) = kf\left(\frac{1}{n}\right)$ for any $k \in \mathbb{Z}^+$ (by using $\frac{1}{n}$ instead of 1 in the above proof). Thus, by taking $k = n$, we get

$$f\left(\frac{1}{n}\right) = \frac{f(1)}{n} = \frac{c}{n}.$$

Therefore, we have $f\left(\frac{m}{n}\right) = mf\left(\frac{1}{n}\right) = c\frac{m}{n}$. So $f(k) = ck$ for all $k \in \mathbb{Q}^+$. Lastly, since $f(-x) = -f(x)$, we have $f(k) = ck$ for all $k \in \mathbb{Q}$. It is not hard to check that all such functions satisfy the condition.

This is a usual way of solving functional equation problems. We usually determine the special values $f(0), f(1)$, etc., and use these to find $f(k)$ for $k \in \mathbb{Z}^+$ or $k \in \mathbb{Z}$. This can often be done by induction. For rational numbers, we write them as $\frac{m}{n}$ and use the values at integers. But such methods cannot be generalized to find the values at all real numbers, since real numbers in general have no arithmetic relations with integers. $\square$

Problem 3.5. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + f(y)) = y + xf(x)$$

for all $x, y \in \mathbb{R}$.

Solution. It is given that

$$f(x^2 + f(y)) = y + xf(x). \tag{1}$$

Putting $x = 0$ in (1), we get $f(f(y)) = y$. To use this fact, we try to generate some terms like $f(f(x))$. For example, we replace x by $f(x)$ in (1). This yields

$$f(f(x)^2 + f(y)) = y + f(x)f(f(x)) = y + xf(x) = f(x^2 + f(y)).$$

Applying f to both sides and using $f(f(t)) = t$, this gives $f(x)^2 + f(y) = x^2 + f(y)$, i.e. $f(x)^2 = x^2$. Therefore, $f(x) = \pm x$ for each x. However, we cannot conclude that the only solutions are $f(x) = x$ and $f(x) = -x$, since there could be some x satisfying $f(x) = x$ and some y satisfying $f(y) = -y$.

We square both sides of (1). Note that

$$\begin{aligned} f(x^2 + f(y))^2 &= (x^2 + f(y))^2 = x^4 + 2x^2 f(y) + y^2, \\ (y + xf(x))^2 &= y^2 + 2xyf(x) + x^2 f(x)^2 = y^2 + 2xyf(x) + x^4. \end{aligned}$$

Equating these and simplifying, we obtain $xf(y) = yf(x)$ for $x \neq 0$. So the signs agree with each other (that means if $f(x) = x$ for some x, then $f(y) = y$ for all y, etc.). Therefore, the only possible solutions are $f(x) = x$ for all $x \in \mathbb{R}$ and $f(x) = -x$ for all $x \in \mathbb{R}$. We can check that both are solutions. $\square$